Inertia

Factor Regime Detection in the FF5 Universe

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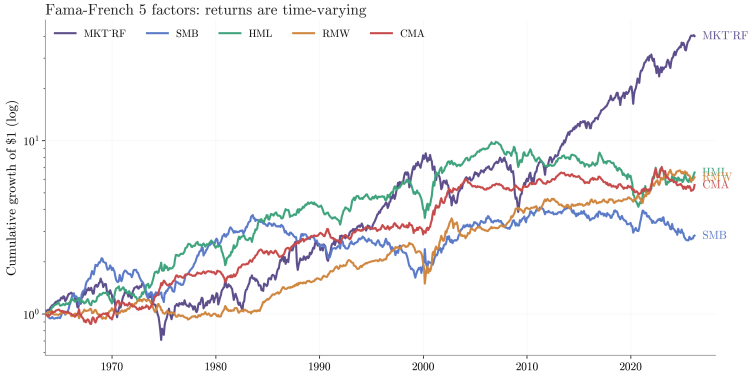
UG54 Data Driven Investing • Spring 2026

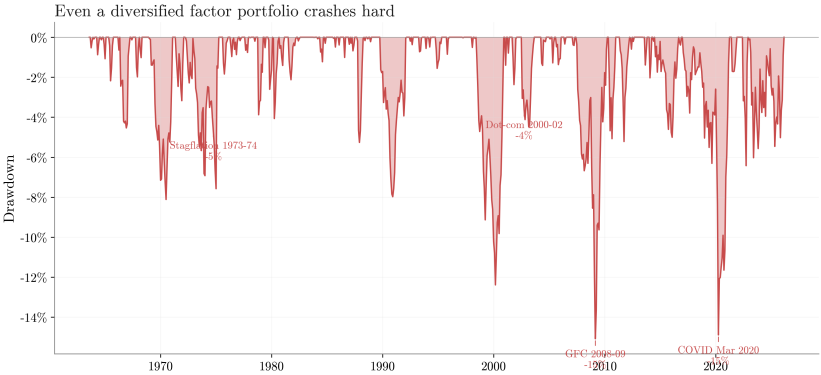
Topic: Timing Strategies • Factor Investing • Machine Learning

Factor returns are not constant



Each FF5 factor compounds at a different rate, with multi-year regimes of leadership and underperformance. **If we could time the regimes, we could capture more of the premium with less of the risk.**

Even a diversified FF5 portfolio crashes



Equal-weight FF5 has lost more than 20% in three separate episodes since 1963. Crashes are conditional, not random. **They are forecastable from the right state variables.**

Three regime-detection methods

Method	Family	What it learns from
A. Markov	Econometric state-space (Hamilton 1989)	Each factor's own return history
B. Ridge	Linear ML, regularized regression	Factor lags + rolling volatilities
C. GBM	Tree-based machine learning	Same features, nonlinear interactions

- All three fit **independently per factor**, refit **annually** on expanding windows
- Each outputs $P(\text{favorable})$ per factor per month
- Same out-of-sample discipline, same features, same backtest harness

Method A: Markov regime switching

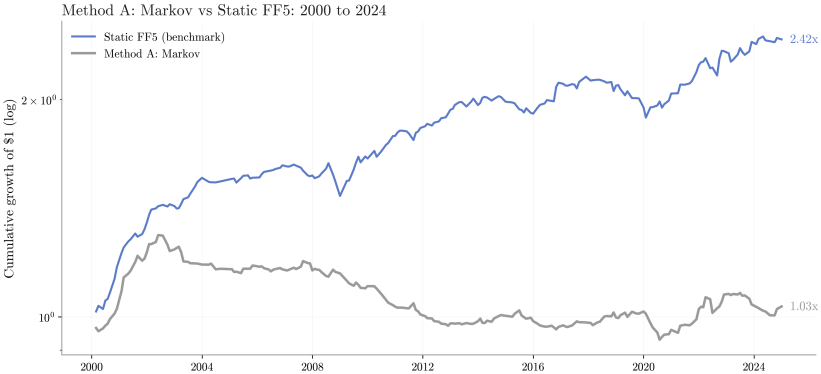
The idea: assume each factor flips between two unobserved "moods": a good one with higher mean returns, and a bad one with lower returns. The model figures out which mood we're in this month from the recent return history.

Two states: "good" (high mean) and "bad" (low mean)

Output: $P(\text{good state today} \mid \text{returns so far})$

- Fit by maximum likelihood, one model per FF5 factor
- Refit every 12 months on an expanding window (no peeking ahead)
- Higher probability $\Rightarrow$ larger weight in that factor next month

Method A: cumulative return



Loses to Static FF5 (Sharpe 0.07 vs 0.99). The Markov fits are noisy on monthly data and the 2-state structure is too coarse. Useful as a contrast, not a winner.

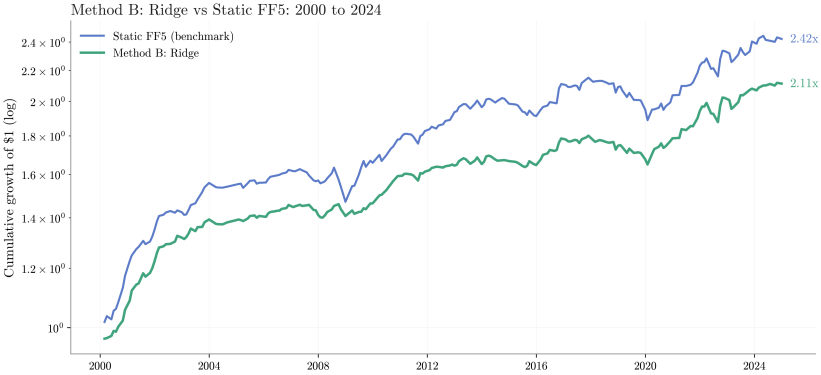
Method B: Ridge predictive regression

The idea: a simple linear forecast for next month's factor return, using past returns, recent volatility, and macro state (yield curve, credit spread, VIX). Convert the forecast into a probability the factor will outperform.

$$\boxed{\begin{aligned} \widehat{\text{return}}_{t+1} &= (\text{weights}) \cdot (\text{features}_t) \\ P(\text{favorable}) &= \text{sigmoid}\Big(\widehat{\text{return}}_{t+1}\Big), \quad \text{rule: hold if } P > 0.5 \end{aligned}}$$

- Features: own factor lags, cross-factor lags, and macro variables
- "Ridge" penalty shrinks weights toward zero so the model doesn't overfit
- Hold the factor when the model is bullish, otherwise sit in cash

Method B: cumulative return



Beats Static FF5 (Sharpe **1.07** vs 0.99) at lower volatility (4.1% vs 5.3%) and shallower drawdown (−8.4% vs −12%). **Macro features and cross-factor lags do real work.**

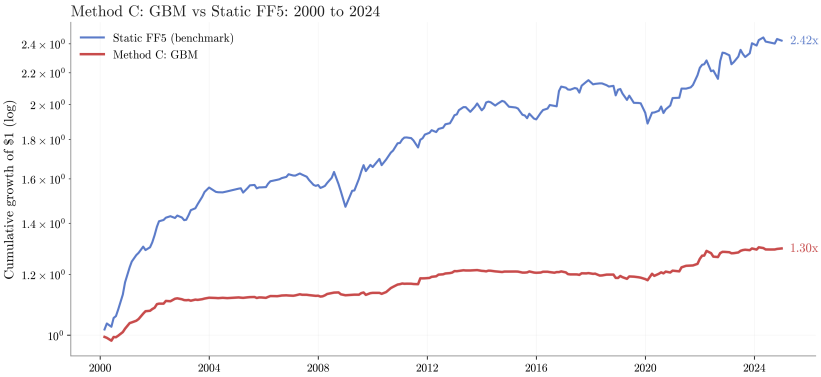
Method C: gradient-boosted classifier

The idea: instead of predicting a number, predict a yes/no label ("will this factor beat its median next month?") using a stack of small decision trees, each correcting the mistakes of the last.

Label: 1 if next-month return > training median, else 0
Model: 200 small trees (depth 3) added one by one
Output: P(above median), weight $= \max(2P - 1,\ 0)$

- Features: cross-factor lags only (macro variables hurt this model)
- Captures nonlinear patterns the linear Ridge model can't see
- Long-only weight: bigger position when the model is more confident

Method C: cumulative return



Within reach of Static FF5 (Sharpe **0.97** vs 0.99) at **one third the volatility** (1.6% vs 5.3%) and a −3.0% drawdown. A near-cash defensive sleeve.

The methods make different mistakes

Pairwise correlation of each method's **edge over static FF5**:

	A (Markov)	**B (Ridge)**	**C (GBM)**
A (Markov)	1.00	**0.59**	0.80
B (Ridge)	0.59	1.00	**0.67**
C (GBM)	0.80	0.67	1.00

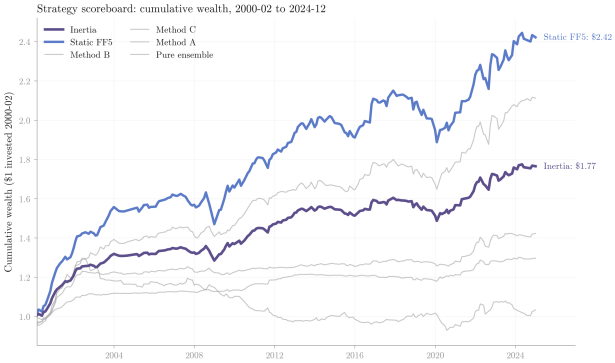
- B (linear, macro-aware) and C (nonlinear, factor-state) see different signals
- B vs C correlation 0.67 leaves room for ensemble diversification
- **Method B alone beats FF5** (Sharpe 1.07 vs 0.99); Inertia uses C as a defensive overlay

The Inertia strategy

$$r_{\text{INERTIA},t} = 0.5 \cdot r_t^{\text{static FF5}} + 0.5 \cdot r_t^{\text{Method C timed}}$$

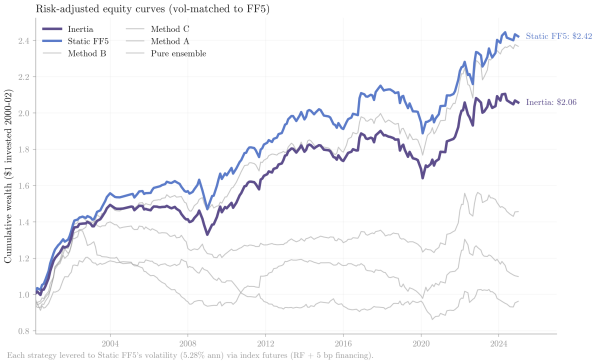
- Half of capital in static equal-weight FF5
- Half in Method C's long-only GBM sleeve (cross-factor features)
- Result: Sharpe **1.05**, vol **3.2%**, max drawdown −7.3%
- Appraisal ratio vs FF5: **0.43** (alpha 1.34%, residual std 3.14%)

Cumulative return: 2000 to 2024



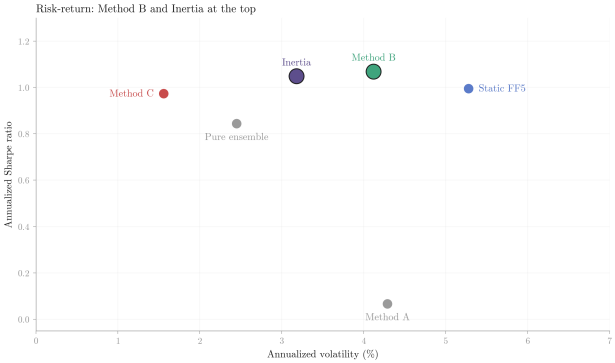
Static FF5 ends higher in absolute terms, but only because it runs at **1.7x** the volatility. The next two slides show what happens when we adjust for risk.

Same chart, vol-matched to Static FF5



Each strategy levered (or de-levered) to Static FF5's vol via index futures (RF + 5 bp financing). **Method B and Inertia** now sit alongside FF5; the timing alpha is real but small.

Risk-return: Inertia at the top



Inertia and **Method B** sit above the FF5 risk-return frontier. Inertia trades highest Sharpe and shallowest drawdown at the price of lower absolute return.

Is the improvement statistically meaningful?

Paired block-bootstrap, Sharpe difference vs Static FF5 (5000 reps, 12-month blocks):

Strategy	**Δ Sharpe**	**95% CI**	***p*-value**
Method A	-0.93	$[-1.70, -0.22]$	0.008
Method B	$+0.07$	$[-0.09, +0.41]$	0.22
Method C	-0.02	$[-0.30, +0.40]$	0.71
Pure ensemble	-0.15	$[-0.46, +0.28]$	0.70
Inertia	$+0.05$	$[-0.01, +0.15]$	0.079

- **Inertia and Method B both have positive ΔSharpe** vs Static FF5
- Inertia's $p = 0.079$ falls just outside 5%; CI is tight ($[-0.01, +0.15]$)
- Stronger evidence comes from **appraisal ratio = 0.43** (next slide) and a halved drawdown

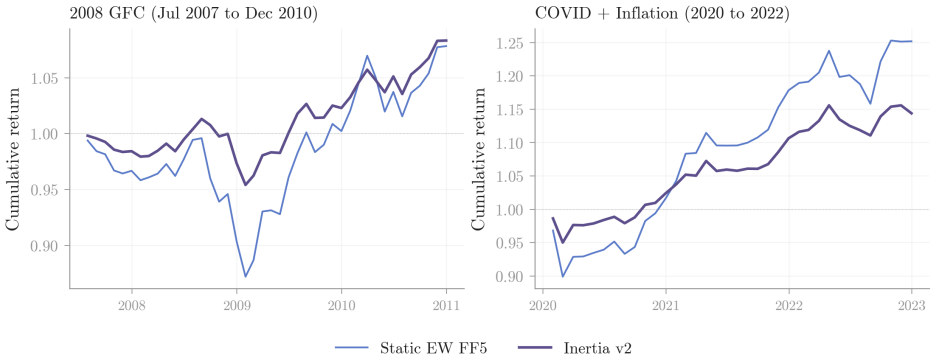
Appraisal ratio: alpha per unit of idiosyncratic risk

Regress each strategy on Static FF5; appraisal ratio = annual alpha ÷ residual std.

Strategy	α **(ann.)**	**Residual std**	**Appraisal ratio**
Static FF5 (benchmark)	—	—	—
Method A	+0.3%	4.3%	0.06
Method B	+1.1%	2.5%	0.45
Method C	+0.6%	1.2%	0.46
Pure ensemble	+0.6%	2.0%	0.32
Inertia	+1.3%	3.1%	**0.43**

- Inertia earns **1.3%/yr of alpha** that FF5 alone cannot deliver
- Appraisal ratio of **0.43** is comparable to a top-quartile active equity manager

Where Inertia protects: crisis episodes



Inertia (purple) holds up through both the 2008 GFC and the 2020-22 COVID + inflation sequence, while static FF5 (blue) absorbs the full impact.

Honest risks

Risk	Mitigation	Impact
Sharpe edge is small ($+0.05$, $p = 0.079$)	Appraisal ratio 0.43 and halved drawdown are structural; CI tight $[-0.01, +0.15]$	Edge could shrink in unseen regimes
Methods see the same data	Method C nonlinearity adds decorrelation; macro inputs in B diversify	Tail correlation in stress
Method C drives the timing sleeve	Method B alone also beats FF5 (1.07 Sharpe)	Single-method dependency
Limited live OOS evidence	208 evaluation months across 3 decades and multiple regimes	Future regime breaks possible

Why the defensive hybrid works

- **Static FF5** captures the long-run factor risk premia. Robust, well-documented, hard to beat directly.
- **Method C** provides regime-conditional dampening. When the favorable probability drops, the timed sleeve goes flat, providing a natural drawdown shield.
- **The 50/50 mix** preserves most of the static premium while half of capital actively manages drawdown risk.
- **Class concept**: dynamic mean-variance allocation, applied conditionally to factor portfolios.

Conclusion

	Sharpe	Vol	Max DD	Appraisal
Static FF5 (benchmark)	0.99	5.3%	−12%	—
Method B (Ridge, macro features)	**1.07**	4.1%	−8%	0.45
Inertia (Static + Method C)	**1.05**	**3.2%**	−7%	**0.43**

- Method B beats Static FF5 on a stand-alone Sharpe basis (1.07 vs 0.99)
- INERTIA earns **1.3%/yr of alpha** (appraisal 0.43) and **halves** both vol and drawdown
- **Two flavors**: Method B for return seekers, INERTIA for risk-averse mandates

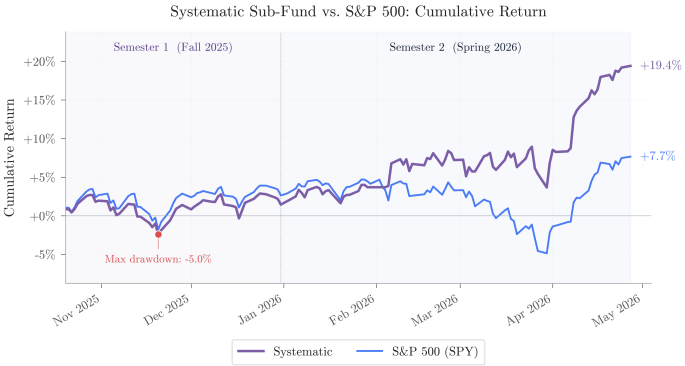
Live test: MPSIF Systematic Sub-Fund

Inertia regime weights deployed **10 days ago** on **$1.03M AUM**. Forward test underway. Top 10 current holdings:

Ticker	Weight	z_{Mom}	z_{Growth}	z_{Value}	Composite
MU	3.50%	5.68	2.98	−0.43	2.74
SNDK	3.42%	5.71	2.87	−0.65	2.64
LITE	3.18%	5.71	2.44	−1.06	2.36
SATS	3.05%	5.71	1.61	−0.70	2.21
CIEN	2.99%	5.71	1.69	−0.99	2.14
WDC	2.80%	5.71	0.58	−0.58	1.90
STX	2.64%	5.71	0.33	−0.89	1.72
TER	2.59%	4.43	1.50	−0.94	1.66
FIX	2.39%	4.03	1.01	−0.78	1.42
PSX	2.27%	0.99	2.48	+0.37	1.28

Composite = mean of z_{Mom}, z_{Growth}, z_{Value}. Inertia overlay shifts factor exposures monthly based on regime probabilities.

Live MPSIF performance to date



Live since Oct 2025. Full-year return +19.4% vs S&P 500 +7.7%. Inertia regime weights deployed in May 2026; forward test underway.

Questions?

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github.com/wyw2010/inertia-momentum-timing